Gate 2: Does Remote Sensing Close the Gap?

Oct 26, 2026 · @Brian Zhou

Summary

Yes, mostly: terrain from elevation data beats the map's tags in regions the model never saw, and learned Tobler matches or beats boosted trees. All five Gate 2 boxes are met; the last one only when learned Tobler uses its own spread.

- Tobler scores 12.5% worse than a refitted slope curve in unseen regions; every learned model scores better than the slope curve.
- Map tags help a little (+0.9% in unseen regions), and terrain beats tags by 0.7 to 2.2 points.
- Learned Tobler's own 90% ranges hold 89.4–89.9% of test segments, so its spread is calibrated.
- Against boosted trees on the same 12 terrain inputs: a tie with the harness's spread, and +0.55 to +0.66 points with its own spread on three of five test setups.

Setup

Every number comes from 289,679 scored 50 m walking segments: 1,992 hikes in 17 regions. The score is CRPS skill against a slope-only curve refitted on each fold's training hikes; 0 means no better than slope alone.

Test setup	What the model never saw	Intervals resample
New hikes	the test hikes	hikes
New hikers	everyone in the test fold	hikers
New 10 km squares	whole 10 km map squares	hikes
Unseen regions	one whole region at a time (17 folds)	regions
Unseen regions and hikers	the region and every hiker who walked there	regions

Intervals are 95% bootstrap intervals over 2,000 draws. Learned Tobler's results average three training seeds.

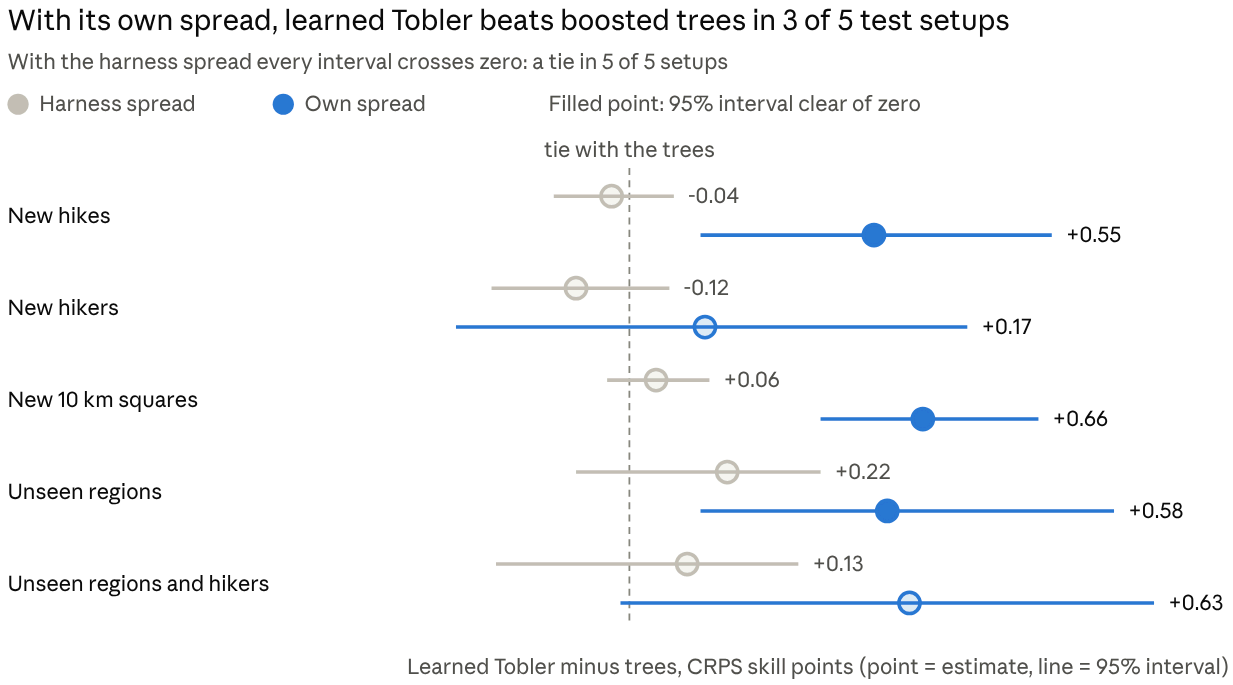
The five boxes

Four boxes are met outright; the fifth, beating plain boosted trees, holds with a caveat.

Gate 2 box (plan §6)	Status	Evidence
Every learned arm beats Tobler on held-out traces	Met	Tobler is 12.5% worse than the slope curve in unseen regions (CRPS). Mean absolute error per segment: Tobler 1.23 km/h, the harness's learned models 1.01–1.06 km/h.
The tags arm beats slope-only	Met	+1.3% / +1.5% on new hikes (additive / trees); +0.9% [0.1, 1.7] / +0.4% [−1.1, 1.6] in unseen regions.
Remote sensing reported against tags, gap quantified	Met	Unseen regions: terrain +2.1% / +2.8%, tags +0.9% / +0.4%. Terrain minus tags +0.7 to +2.2 points; terrain wins in 16 of 17 regions (additive). It holds with the region's hikers held out too.
Predicted variances calibrated, not just the means	Met	Learned Tobler's own 90% ranges hold 89.4–89.9% of test segments in every setup.
Your model beats plain boosted trees on the same features	Met, with a caveat	A tie with the harness's spread (−0.12 to +0.22 points). With its own spread, +0.55 to +0.66 points on three setups, intervals above zero. The trees had no learned spread.

Learned Tobler vs boosted trees

Learned Tobler ties the trees with the harness's spread and beats them in three of five setups with its own spread.



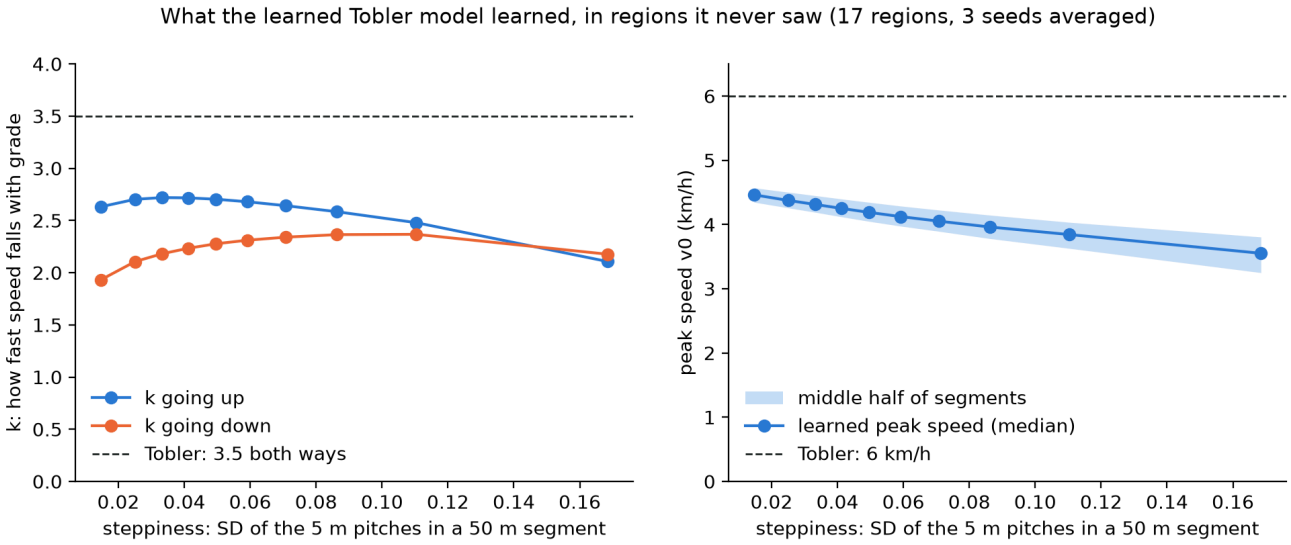
results/t3_learned_tobler/gate2_summary.csv · 5 test setups, 3 seeds averaged, 2,000 bootstrap draws

Test setup	Trees	Learned Tobler, harness spread	Learned Tobler, own spread	Own 90% range holds
New hikes	+3.22%	+3.19%	+3.77%	89.8%
New hikers	+3.19%	+3.07%	+3.36%	89.9%
New 10 km squares	+2.83%	+2.90%	+3.50%	89.7%
Unseen regions	+2.79%	+3.01%	+3.37%	89.4%
Unseen regions and hikers	+2.42%	+2.55%	+3.05%	89.4%

Skill is the improvement over slope alone; the trees are the harness's, refitted on the same folds. Region by region, learned Tobler beats the trees in 13 of 17 unseen regions and slope alone in all 17.

What the model learned

On smooth trail the learned k is 2.6 going up and 1.9 going down, against Tobler's 3.5. On the steppiest trail it is about 2.1–2.2 both ways. Steppy trail is slow at every grade: the peak speed falls from 4.5 to 3.6 km/h, against Tobler's 6.



Each point is the median of a tenth of all segments, each scored while its region was held out (17 regions, three seeds averaged). Steppiness is the spread of the 5 m pitches inside a 50 m segment: about 0.015 on smooth trail, 0.17 on rock steps.

How learned Tobler works

It keeps Tobler's shape but lets a small network set the constants for each 50 m segment from that segment's terrain.

```
\ln v = a(x) - k_{\pm}(x)\left(\sqrt{\left(s + s_0(x)\right)^2 + e^2} - e\right)
```

- s is the slope; x is the segment's 12 terrain measures from the 1 m and 5 m elevation models: steppiness, steepest pitch, hidden climbing, relief along the path, roughness and side slope.
- A network with two hidden layers of 32 outputs, for every segment, the log peak speed a, the drop-off k going up and going down, the peak position s0, and a spread σ. The rounding e stays fixed.
- It starts exactly on the refitted slope curve (last-layer weights zero, reference constants as biases), so training only learns departures from it.

- Training: Student-t likelihood with ν = 5 (robust to GPS glitches), Adam, batches of 2,048 segments, early stopping on held-out hikes. Inputs are clipped to the training data's 0.1st–99.9th percentiles.
- Scoring, two ways: with the harness's spread (residuals on calibration hikes, as for every Table I model), and with its own Student-t spread.

Caveats

The tie with the trees is solid; the win depends on learned Tobler's own spread, which the trees did not get.

- **No learned spread for the trees.** An earlier test gave them a segment-dependent spread and it did not help (decision #98), so the comparison is fair but not exhaustive.
- **Elevation glitches.** The first full run broke on them: hidden climbing sat 524 standard deviations above its mean in one segment. Trees ignore such values; the network extrapolated them, so its inputs are now clipped. The glitched segments need checking in the next data release.
- **Seeds.** Across three seeds the harness-spread score moves by up to 0.24 points within a setup, and the learned constants by about 0.1, so the figure averages the seeds.
- **North Shore.** Its 90% ranges hold only 81.9%, as they do for every model (decision #115).
- **Data version.** Everything uses the frozen T1 v1 data, like the paper.

Next steps

With Gate 2 met, the plan moves to imagery and the full paper draft.

1. Week 12 (9–15 Nov): the NAIP aerial-imagery arm if there is time, feeding the same model; the IGARSS template once the submission site opens on Nov 10, 2026.
2. Week 13 (16–22 Nov): the full paper draft in my own words by Nov 22, 2026, with learned Tobler as a new row and the steppiness figure.

Where to find everything

All code, results and figures are in the private GitHub repo Brian24NX/GeoAI.

- Gate 2 numbers (the table and chart above): travel-speed-gate1/results/t3_learned_tobler/gate2_summary.csv

- Steppiness figure: travel-speed-gate1/exercises/week10_learned_tobler/fig_constants.png
- Week 9 log-linear check: travel-speed-gate1/exercises/week09_loglinear/fig_loglinear.png
- Learned-Tobler model and the Gate 2 runs: travel-speed-gate1/exercises/week10_learned_tobler/ (`learned_tobler.py`, `run_all.py`, `gate2_summary.py`, `plot_constants.py`)
- Paper draft 0: paper/main.pdf
- Week 12 imagery plan: docs/T3_naip_imagery_design.md
- Lab-meeting slides (23 Oct): meeting_plots/oct23_lab/Zhou_LabMeeting_23Oct.pdf
- Every number and decision: DECISIONS_LOG.md (#95, #115, #116, #120–#122); weekly progress: PROGRESS.md

Code and drafts were developed with AI assistance (Claude).